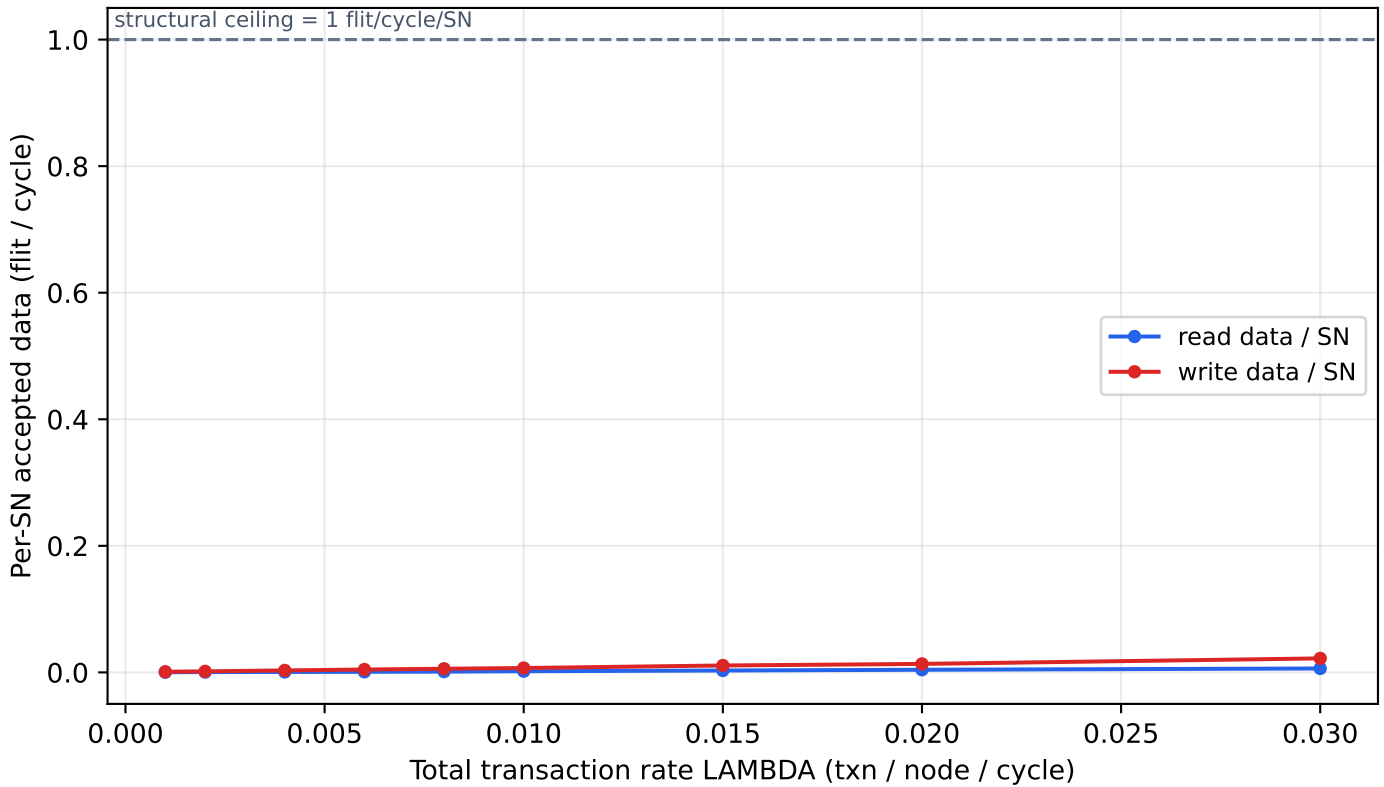
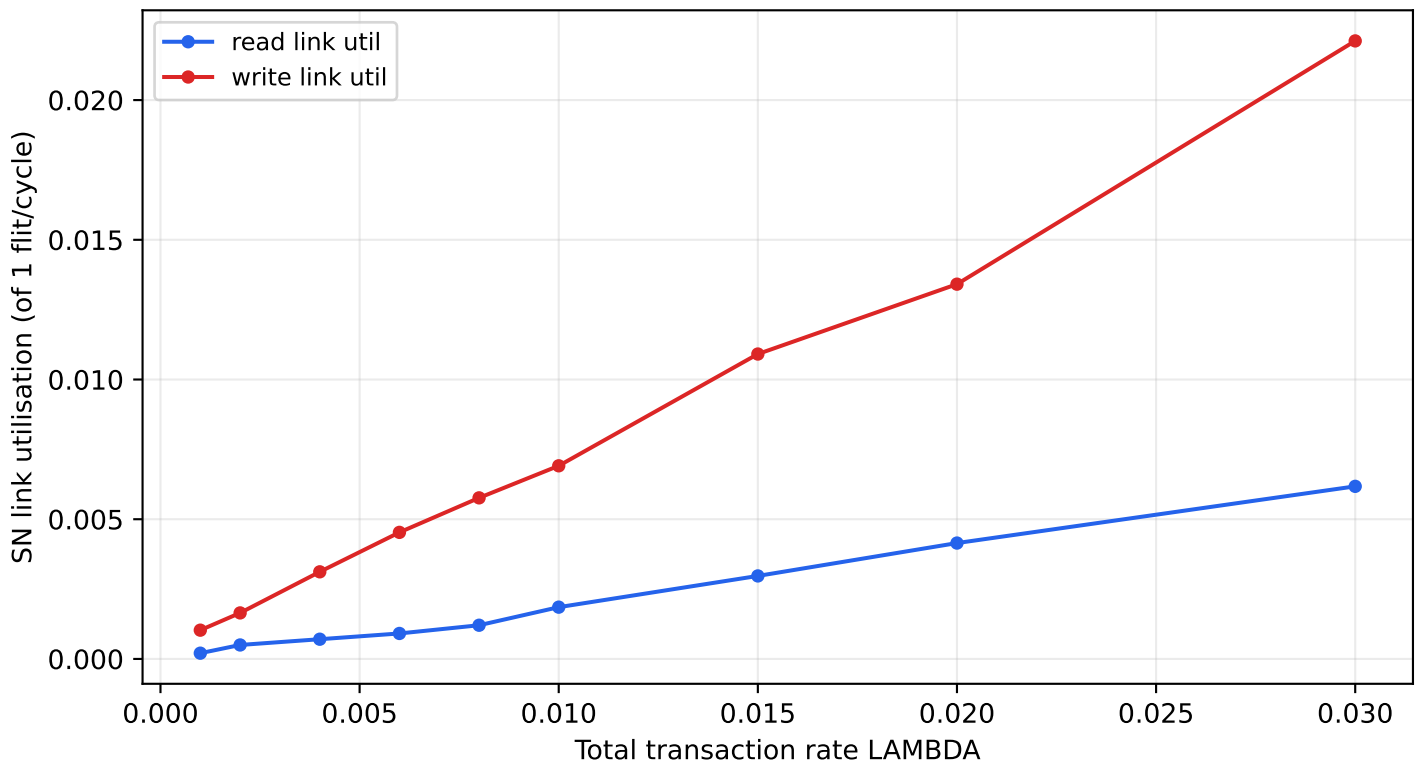


SN (DDR) Throughput - Current Traffic Mix 6x7 mesh + 4 SN, XY routing, 2-cycle links

Per-SN read/write data throughput vs offered load



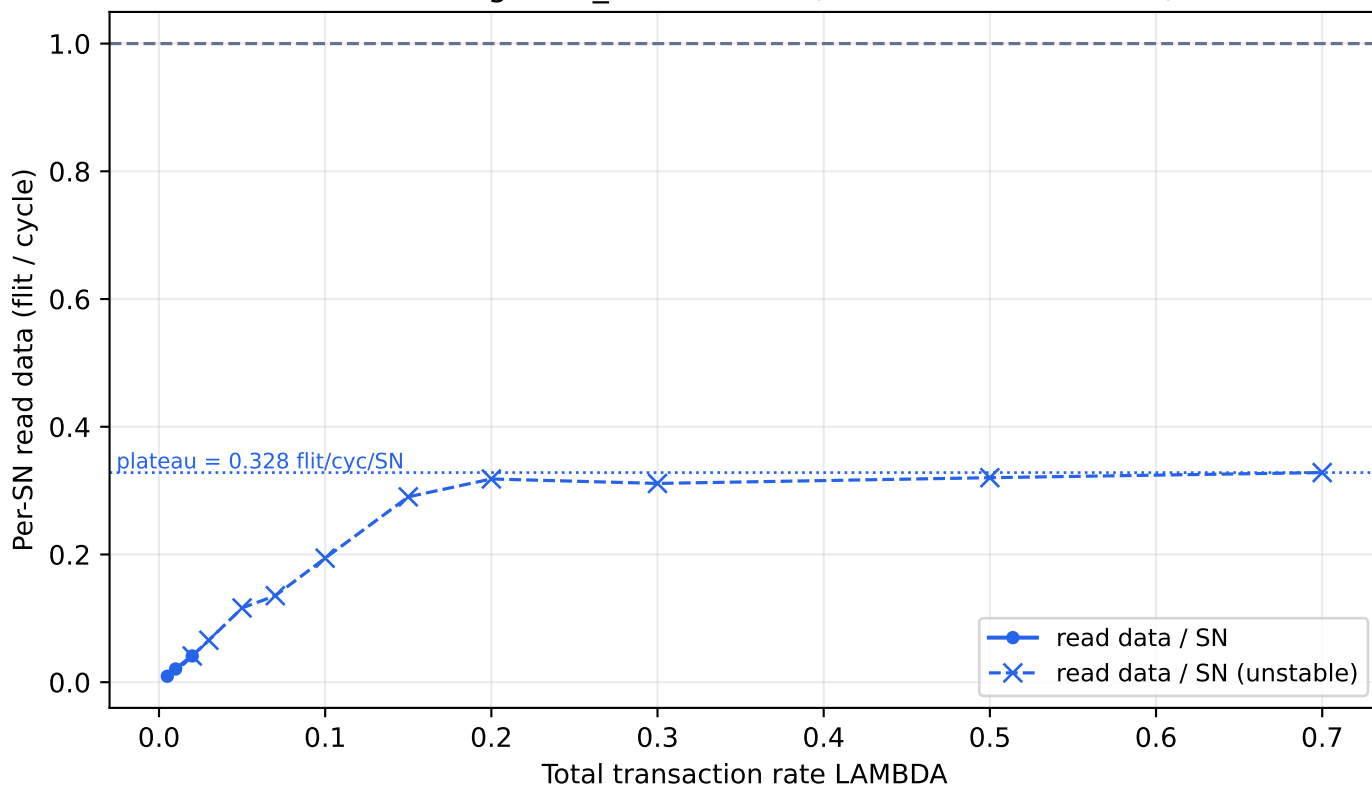
SN terminal-link utilisation (current mix)



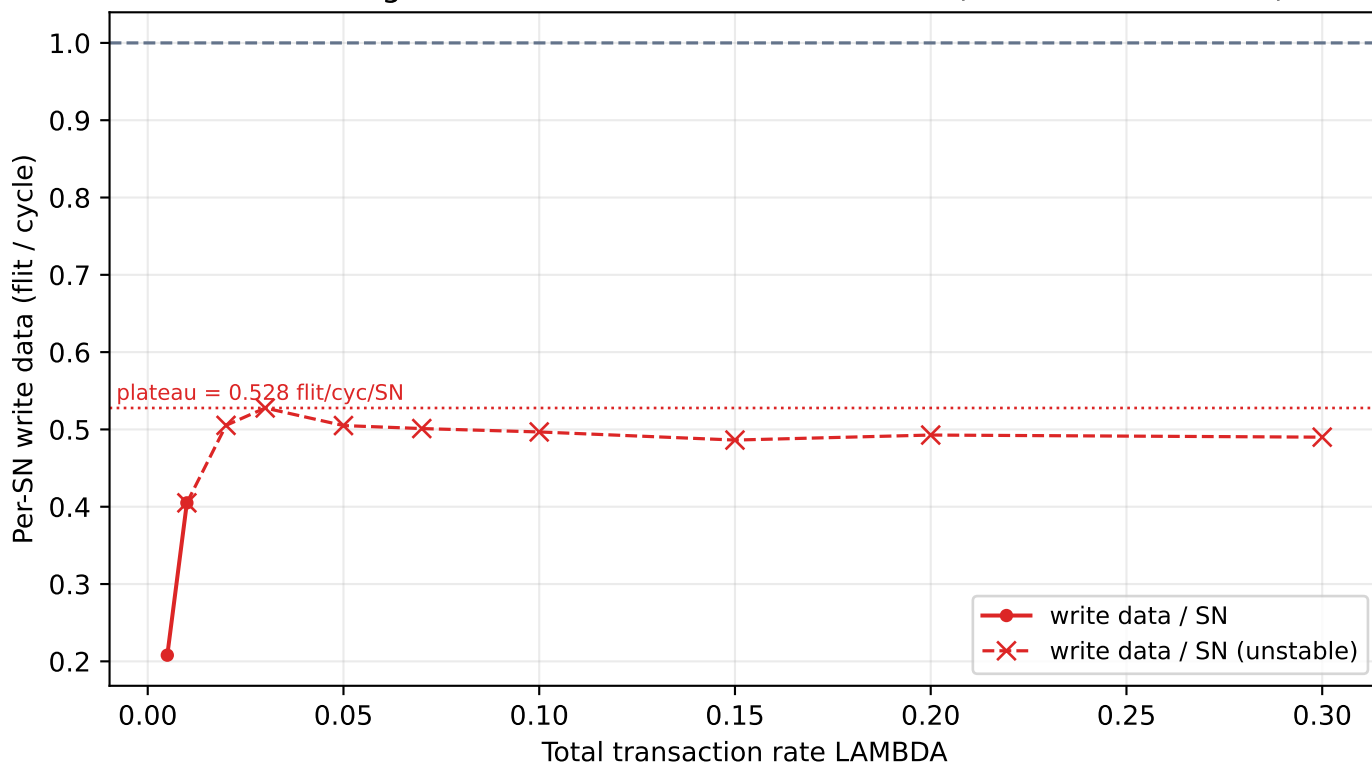
At the network saturation point the SN links carry only ~0.62% (read) / ~2.21% (write) of their 1 flit/cycle capacity: under the current mix the SN is far from being the bottleneck.

SN Read / Write Throughput Ceiling (focused mixes)

Read ceiling: dmt_miss=100% (all read data from SN)



Write ceiling: WriteBack + L3EvictToSN=100% (all write data to SN)



SN Throughput Ceiling - Summary

Metric	per SN (flit/cyc)	4-SN total (flit/cyc)	util vs 1 flit/cyc	per-SN bytes/cyc
Read ceiling (SN->RN)	0.328	1.313	32.8%	5.25 B/cyc
Write ceiling (HN->SN)	0.528	2.111	52.8%	8.44 B/cyc
Read @ current mix (sat.)	0.0062	0.0247	0.62%	0.10 B/cyc
Write @ current mix (sat.)	0.0221	0.0885	2.21%	0.35 B/cyc

- Structural ceiling: each SN attaches to its router with one full-duplex terminal link = 1 flit/cycle per direction (channel_width is power-only; no crossbar speedup). Read data (SN->RN) and write data (HN->SN) use separate links, each capped at 1 flit/cycle/SN -> 4 flit/cycle aggregated over the 4 SNs.

- Read ceiling: with dmt_miss=100% all read data is served by the 4 SNs; each SN injects CompData_DMT (2-flit) on the DAT subnet. As LAMBDA rises the SN DAT inject link is the binding constraint and the per-SN read throughput plateaus (unstable points captured from the last periodic snapshot).

- Write ceiling: with WriteBack + L3EvictToSN=100% all writeback data lands on the 4 SNs; each SN ejects L3EvictData (2-flit) on DAT, plateauing at its ejection-link limit.

- Gap vs 1 flit/cycle reflects the router microarchitecture: 2 VCs/input, 2-flit VC buffers, single alloc iteration, output-first round-robin, and 2-cycle inter-router links (deeper VC buffers raise the achievable fraction).

- Under the current mix the SN links sit at <1% utilisation even at network saturation, so the SN is not today's bottleneck; the ceiling above is what a memory-bound workload could extract from this topology.